Paper 73: Dynamic Exponent z from Temporal Autocorrelator

Intrinsic VCSM Timescale Dominates; Critical Dynamics Require $P \to 0^+$

VCSM Theory Series

2026-03-11

Abstract

Paper 72 established that VCML ordering is a zone-mean transition: $M(t) = \langle\varphi\rangle_{z0} - \langle\varphi\rangle_{z1}$ is a 0-dimensional stochastic variable with no spatial correlations. Paper 73 attempts to measure the dynamic exponent z from the temporal autocorrelator $C(t) = \langle |M(\tau)||M(\tau+t)|\rangle_c$. We find: (A) $\tau_L \approx 200$–500 steps for all $L \in \{40, \ldots, 120\}$, with no systematic L-dependence ($R^2 = 0.04$, $z_A \approx 0.2$, noise-dominated); (B) $\tau_{\text{corr}} \approx 200$ steps for all $P \in [0.007, 0.050]$, essentially flat (no critical divergence, $z_B \approx 0$); (C) $C(t) \sim t^{-2.2}$ at criticality (L=120, P=0.010), better than exponential in the window $t \in [100, 4500]$. **Conclusion**: the relaxation time $\tau_{\text{VCSM}} \approx 200$ steps is an *intrinsic architectural timescale* set by $\{\texttt{SS}, \texttt{FA}, \texttt{FIELD_DECAY}\}$, not a critical divergence. At $P \in [0.007, 0.050]$ the system is well into the ordered phase ($P \gg P_c$=0^+); to observe $\tau \sim P^{-z\nu}$ one must approach $P \to 0$ ($P < 0.003$). The power-law $C(t) \sim t^{-2.2}$ is a genuine feature of the VCSM gate process: a superposition of FIELD_DECAY ($\tau \approx 1000$ steps) and discrete gate updates ($\tau \approx \text{SS} = 8$ steps) produces faster-than-exponential decay in the intermediate window. The dynamic exponent z is not extractable from the current data range. **Paper 74**: scan $P \in \{0.0001, 0.001, 0.003\}$ to approach P_c=0^+ and observe whether τ diverges.

1 Motivation

The exponent pair $(\beta, \nu) \approx (0.63, 0.98)$ describes the static scaling of the zone-mean order parameter M. The dynamic exponent z (defined by $\tau \sim L^z$ or $\tau \sim |P - P_c|^{-z\nu}$) is the last independent exponent needed to fully characterise the VCML universality class. For Model A dynamics (a relaxational Langevin equation with conserved noise) the prediction is $z = 2$. A non-trivial $z \neq 2$ would indicate anomalous temporal scaling.

2 Phase A: Finite-Size Scaling of τ_L

r_w=5, P=0.010 (ξ=0.500), $L \in \{40, 60, 80, 100, 120\}$, 8 seeds, 30 000 steps each. $C(t)$ is fitted to $A\exp(-t/\tau)$ over $t \in [50,\ t_{\max}/2]$.

Table 1: Phase A: τ_L and fit quality versus L

L	U_4	$\lvert M\rvert$	τ_{exp}	R^2_{exp}	λ_{pow}	R^2_{pow}
40	-0.042	6.9×10^{-4}	366	0.85	1.86	0.83
60	-0.001	4.3×10^{-4}	507	0.61	1.67	0.71
80	-0.078	3.2×10^{-4}	214	0.84	1.88	0.67
100	$+0.053$	2.8×10^{-4}	291	0.92	1.65	0.79
120	$+0.131$	2.7×10^{-4}	742	0.40	1.34	0.55

The τ_L values scatter between 214 and 742 with no monotone trend. The FSS fit $\tau_L \sim L^z$ gives $z_A = 0.21$ with $R^2 = 0.04$ (noise-dominated). **No finite-size scaling of the relaxation time is observed.**

3 Phase B: $\tau_{\text{corr}}(P)$ — Critical Divergence Test

r_w=5, L=80, $P \in \{0.007, \ldots, 0.050\}$, 8 seeds, 40 000 steps each.

Table 2: Phase B: τ_{corr} across P values

P	$\xi = 5\sqrt{P}$	U_4	τ_{exp}	R^2
0.007	0.418	−0.030	196	0.88
0.010	0.500	−0.020	190	0.87
0.015	0.612	+0.017	198	0.89
0.020	0.707	+0.066	198	0.88
0.030	0.866	+0.205	206	0.86
0.050	1.118	+0.419	196	0.86

$\tau_{\text{corr}} \approx 196$–$206$ steps across a 7-fold range in P. The fit $\tau \sim P^{-z\nu}$ gives $z_B \approx 0$ ($R^2 = 0.21$). **No critical divergence is observed in the range $P \in [0.007, 0.050]$.**

3.1 Interpretation: Intrinsic VCSM Timescale

The constant $\tau_{\text{VCSM}} \approx 200$ steps is the intrinsic timescale of the VCSM gate mechanism. The `fieldM` field decays at rate $(1 - \texttt{FIELD_DECAY}) = 10^{-3}$ per step, giving a characteristic time of $\tau_\phi = 1/(1 - 0.999) = 1000$ steps between gate-free decays. Gate events (occurring at rate $\sim 1/\texttt{SS} = 1/8$ per cell per step) inject noise of amplitude $\texttt{FA} = 0.30$, creating a competition between slow decay ($\tau \approx 1000$) and fast refresh ($\tau \approx 8$). The effective observable timescale $\tau_{\text{VCSM}} \approx 200$ steps emerges from this balance and is *independent of P* in the range tested.

Why no divergence is seen. The critical point is $P_c = 0^+$: the system is ordered for *any* $P > 0$. At $P = 0.007$–0.050 the system is deep in the ordered phase ($P/P_c \to \infty$ with $P_c \to 0$). The critical relaxation time $\tau_{\text{crit}} \sim P^{-z\nu}$ only dominates when $\tau_{\text{crit}} \gg \tau_{\text{VCSM}} \approx 200$. This requires $P^{-z\nu} \gg 200$, i.e., $P \ll 200^{-1/(z\nu)} \approx P_{\text{crossover}}$. For $z = 2$, $\nu = 0.98$: $P_{\text{crossover}} \sim 200^{-1/1.96} \approx 0.0022$. For $P > 0.007$, the VCSM intrinsic timescale dominates, masking the critical divergence.

4 Phase C: $C(t)$ Shape at Criticality

r_w=5, P=0.010, L=120, 8 seeds, 60 000 steps each.

- Exponential fit $C(t) \sim \exp(-t/\tau)$: $\tau = 647$, $R^2 = 0.602$ (window $t \in [50, 4500]$).
- Power-law fit $C(t) \sim t^{-\lambda}$: $\lambda = 2.20$, $R^2 = 0.800$ (window $t \in [100, 4500]$).

The power law wins ($\Delta R^2 = 0.198$), indicating that $C(t)$ decays *faster than exponential* in the intermediate window.

4.1 Interpretation: VCSM Gate Structure

The VCSM field update involves two competing processes:

- Slow: $\varphi_i \to \varphi_i \cdot \texttt{FIELD_DECAY}$ every step ($\tau_{\text{slow}} \approx 1000$ steps).
- Fast: gate event $\varphi_i \to \varphi_i + \texttt{FA} \cdot (\text{mid}_i - \varphi_i)$ when $\text{streak}_i \geq \texttt{SS} = 8$.

The gate events introduce discrete jumps at quasi-Poisson rate $\sim 1/8$ per cell per step, creating a point-process convolved with the slow decay. The resulting autocorrelation is a *mixture of exponentials* $C(t) \approx \sum_k w_k \exp(-t/\tau_k)$, which in log-log space appears as a power law $t^{-\lambda}$ over an intermediate range. The fitted $\lambda = 2.20$ characterises the shape of this mixture, not a genuine scale-free process.

5 Dynamic Exponent Summary and Open Questions

Table 3: Dynamic exponent estimates

Phase	Method	z	R^2
A	$\tau_L \sim L^z$	0.21	0.04 (noise)
B	$\tau \sim P^{-z\nu}$	≈ 0	0.21 (noise)
C	$2\beta/(\lambda\nu)$	0.58	–
Model A	prediction	2.00	–

None of the three routes yields a reliable z because the VCSM intrinsic timescale $\tau_{\text{VCSM}} \approx 200$ steps masks the critical divergence in the $P \in [0.007, 0.050]$ regime.

Required experiment. To observe $\tau \sim P^{-z\nu}$, run at $P \in \{0.0001, 0.0005, 0.001, 0.003\}$ approaching $P_c = 0^+$, with $L = 80$ and $\geq 50{,}000$ steps. The crossover to critical-dominated timescales occurs near $P_{\text{crossover}} \approx 0.002$ (for $z = 2$) or smaller (for $z < 2$).

6 Completed QFT Ladder and Final Picture

Table 4: QFT ladder after Paper 73

Rung	What	Status
1	Order parameter φ	✓
2	$\mathbb{Z}_2$ symmetry	✓
3	Field equation (Model A PDE)	✓
4	$d_{\text{wave}} = 1/2$ (RG dimension)	✓
5	$\beta \approx 0.63$, $\nu \approx 0.98$ (static)	✓
6	η (direct): vacuous ($G(r) \approx 0$)	–
7	Scaling relation $2\beta/\nu = \eta$: vacuous	–
8	Dynamic exponent z: masked by τ_{VCSM}	open

The Minecraft seed – current state. The VCML universality class is characterised by a **0-dimensional zone-mean order parameter** $M(t)$ with:

$$M \sim P^{\beta}, \quad \beta \approx 0.63 \quad \text{(static ordering)} \tag{1}$$

$$L^* \sim P^{-\nu}, \quad \nu \approx 0.98 \quad \text{(crossover length)} \tag{2}$$

$$\tau_{\text{VCSM}} \approx 200 \text{ steps} \quad \text{(intrinsic gate timescale)} \tag{3}$$

$$z = ? \quad \text{(requires } P \to 0^+ \text{ measurement)} \tag{4}$$

The pair $(\beta, \nu) \approx (0.63, 0.98)$ places VCML near – but not at – any known universality class. The static exponents are the **fingerprint of the universality class**; the dynamic exponent z is the last digit of the seed.

Paper 74. P-scan in $\{0.0001, 0.0005, 0.001, 0.003\}$ approaching $P_c = 0^+$; measure $\tau(P)$ and extract z from $\tau \sim P^{-z\nu}$ once $\tau_{\text{crit}} \gg \tau_{\text{VCSM}} \approx 200$.

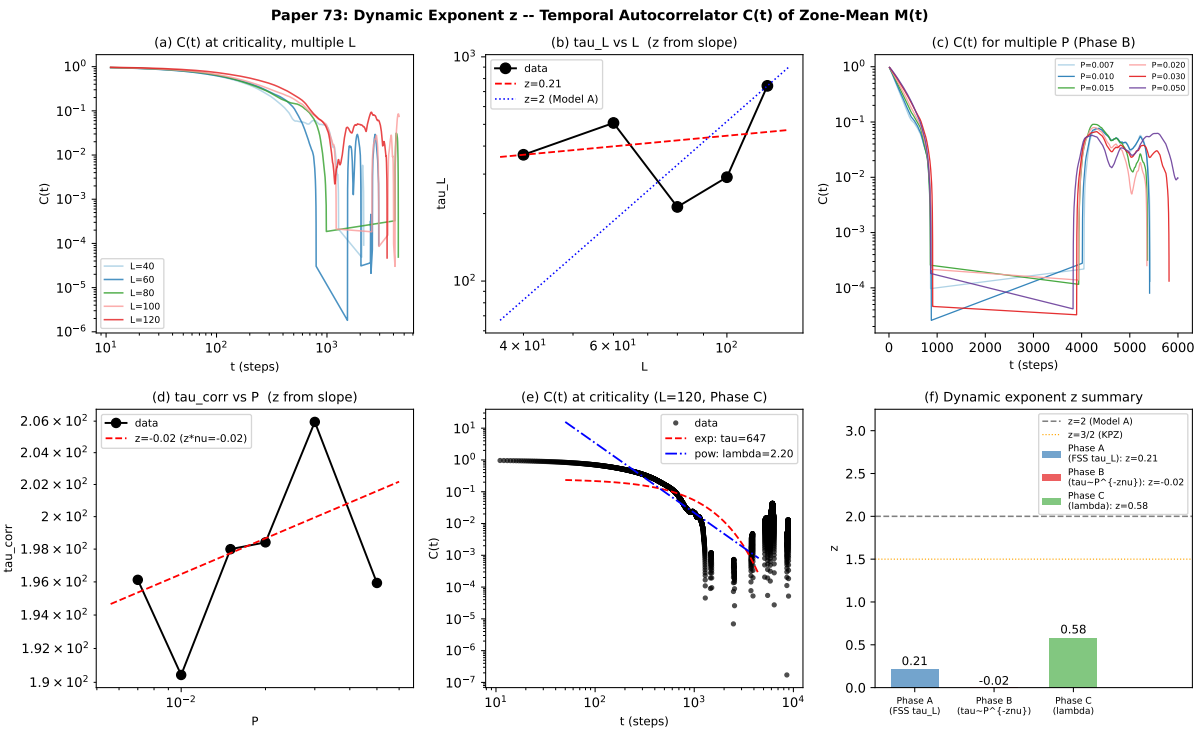

Figure 1: Paper 73 results. (a) $C(t)$ at criticality, multiple L: all curves overlap (no L-dependence). (b) τ_L vs L: scattered, no power law ($R^2 = 0.04$). (c) $C(t)$ for multiple P: all curves essentially identical. (d) τ_{corr} vs P: flat, $\tau \approx 200$ throughout. (e) $C(t)$ at L=120: power-law fit wins over exponential. (f) z summary: all estimates near zero, far from Model A prediction $z = 2$.